

CSE 4345: Big Data Analytics

Week 6, Part 1 — Apache Spark: RDDs, DataFrames & In-Memory Computing

Department of Computer Science & Engineering
Premier University, Chittagong

Semester 2026

Course & Lecture Information

Lecture Identity

Course: CSE 4345
Week / Part: 6 of 11 | Part 1 of 2
CLO: CLO3
PLO: PLO(e), PLO(f)

CLO3

“Explain big data techniques, tools, and technologies used to store, manage, and analyse large datasets”

Course Progression

Week 4: MapReduce, Pig, Hive, HBase
Week 5: YARN, file formats, compression

Part 1 Covers

- Why MapReduce fails for iterative algorithms
- Spark history and the in-memory insight
- Spark architecture: Driver, Executors, Cluster Manager
- RDD: the core abstraction (immutable, partitioned, lineage)
- RDD lineage and fault-tolerant recomputation
- Transformations vs. actions; lazy evaluation
- Narrow vs. wide dependencies; stages and tasks
- PySpark programming examples
- Spark SQL and the DataFrame API
- Spark ecosystem (MLlib, Structured Streaming)

Lecture Agenda

- 1 The MapReduce Bottleneck
- 2 Apache Spark: Architecture & RDDs
- 3 Transformations, Actions, and Lazy Evaluation
- 4 PySpark Programming
- 5 Spark SQL and DataFrames
- 6 The Spark Ecosystem
- 7 MapReduce vs. Spark: Head-to-Head
- 8 Ivy League Alignment
- 9 Student Assessment Pool

The MapReduce Bottleneck

Why MapReduce Fails for Iterative Algorithms

The Fundamental Problem

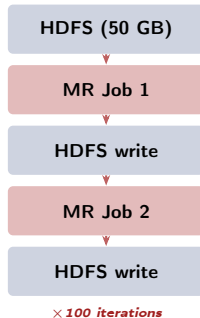
MapReduce writes **every intermediate result to HDFS** before the next step can begin. For algorithms that must pass over the same data repeatedly (machine learning, graph algorithms), this incurs enormous disk I/O cost.

K-Means Clustering: 100 Iterations

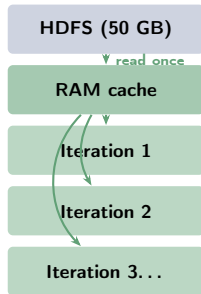
Each iteration: read training data → compute new centroids → write to HDFS.

- Dataset: 50 GB on HDFS
- **MapReduce**: 100 iterations $\times$ (50 GB read + 50 GB write) = **10 TB disk I/O**
- **Spark**: iteration 1 reads 50 GB from HDFS and caches in RAM. Iterations 2..100 read from memory.

MapReduce (slow)



Spark (fast)



Apache Spark: Architecture & RDDs

Spark History and the In-Memory Insight

Origin: UC Berkeley AMPLab (2009–2013)

- **2009:** Matei Zaharia, PhD student at UC Berkeley, observes MapReduce is inefficient for iterative ML and interactive queries
- **2010:** First Spark paper (HotCloud 2010) — “Cluster Computing with Working Sets”
- **2012:** RDD paper (NSDI 2012) formalises fault tolerance via **lineage** — Best Paper Award
- **2013:** Spark donated to Apache Software Foundation; Databricks founded
- **2014:** Spark sets world record in 100 TB sort benchmark (23 min vs. 72 min for Hadoop)
- **2016–present:** Default big data engine at Netflix, LinkedIn, Alibaba, Uber

The Core Insight

*“We introduce the concept of **re-silient distributed datasets** (RDDs), a distributed memory abstraction that lets programmers perform in-memory computations on large clusters in a fault-tolerant manner.”*

— Zaharia et al., NSDI 2012

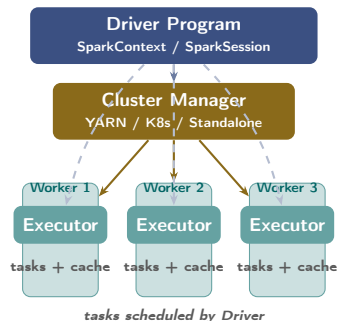
Why It Matters for Bangladesh

Bangladesh's largest ML use cases (fraud detection at bKash, disease prediction at DGHS, ride-demand forecasting at Pathao) all involve iterative algorithms over large datasets. These are **exactly** the workloads where Spark outperforms

Spark Architecture

Three-Tier Architecture

- 1 **Driver Program:** runs the user's `main()` function; creates `SparkContext`; builds the DAG of transformations; coordinates execution
- 2 **Cluster Manager:** allocates resources across the cluster (YARN, Kubernetes, Mesos, or Spark Standalone)
- 3 **Executors:** JVM processes launched on Worker Nodes; execute tasks; store cached RDD partitions in memory



SparkContext vs. SparkSession

In Spark 1.x, `SparkContext` was the entry point. In Spark 2.x+, `SparkSession` unifies SQL, DataFrames, and RDDs under one entry point. `SparkContext` is still

Resilient Distributed Datasets (RDDs)

Definition (Zaharia et al., 2012)

An RDD is a **read-only, partitioned collection of records** that can only be created by:

- 1 Transforming an existing RDD (e.g., map, filter)
- 2 Loading data from stable storage (e.g., HDFS, S3)

Five key properties:

- **Immutable:** never modified in place; transformations create new RDDs
- **Partitioned:** split across Executors for parallel processing
- **Lazy:** transformations only execute when an action is called

What “Resilient” Actually Means

Traditional fault tolerance = **replicate data** (3 copies, like HDFS).

RDD fault tolerance = **re-derive data** from its lineage (the computation graph).

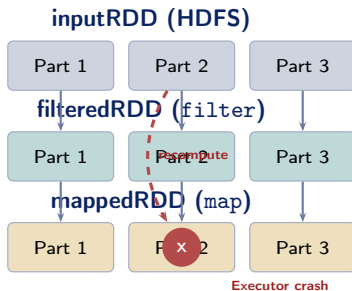
If Partition 3 of an RDD is lost:

- 1 Spark reads the original HDFS input for that partition
- 2 Re-applies the same sequence of transformations
- 3 Result: the lost partition is regenerated

No data replication overhead. The computation is the backup.

The Key Trade-off

RDD Lineage: The Fault-Tolerance Mechanism



Recovery Path

1. Spark detects lost Part 2
2. Traces lineage: mappedRDD was derived from filteredRDD
filteredRDD was derived from inputRDD
3. Re-reads inputRDD Part 2 from HDFS
4. Re-applies filter then map
5. Part 2 restored **without** restarting the whole job

Transformations, Actions, and Lazy Evaluation

Transformations vs. Actions

Transformations (Lazy)

Build the **computation DAG** but do not execute immediately. Return a new RDD.

Transformation	What it does
----------------	--------------

map(f)	Apply f to each element
filter(f)	Keep elements where f is True
flatMap(f)	Map then flatten result
distinct()	Remove duplicates
reduceByKey(f)	Aggregate values per key
join(other)	Inner join on key
groupByKey()	Group values by key
cache()	Persist RDD in memory

Actions (Eager)

Trigger execution of the DAG. Return a value to the Driver or write to storage.

Action	What it returns
--------	-----------------

collect()	All elements (list)
count()	Number of elements
first()	First element
take(n)	First n elements
reduce(f)	Single aggregate value
saveAsTextFile	Write to HDFS/disk

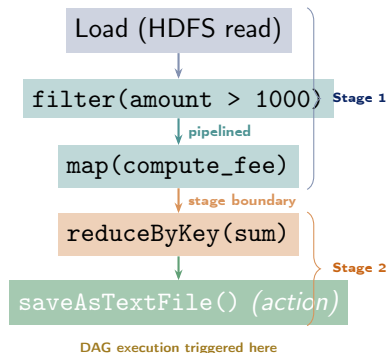
Rule:

Lazy Evaluation: Why It Matters

What Lazy Evaluation Means

When you call `rdd.filter(f).map(g)`, Spark does **nothing** yet. It only records the computation graph (DAG). When you call `.count()` or `.collect()`, Spark:

- 1 Analyses the full DAG
- 2 Applies the **Catalyst optimizer** (for DataFrames) or Spark's DAG scheduler
- 3 **Pipelines** compatible transformations: runs `filter` and `map` in a single pass over each partition — never writing intermediate results to disk
- 4 Executes only the partitions needed



Narrow vs. Wide Dependencies: Stage Boundaries

Narrow Dependencies (no shuffle)

Each output partition depends on **at most one** input partition. Can be **pipelined** within a single stage.

- map, filter, flatMap
- union, mapPartitions
- coalesce (reduce partitions, no shuffle)

Recovery: if a partition is lost, recompute only that partition from its single parent partition.

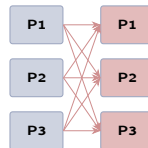
Narrow (map)



1-to-1 mapping

one stage

Wide (groupByKey)



all-to-all (shuffle)

two stages

Wide Dependencies (requires shuffle)

Each output partition may depend on **multiple** input partitions. Data must move across the network (shuffle). Creates a **stage boundary**.

Performance Tip

Prefer `reduceByKey` over `groupByKey`: `reduceByKey` applies a local combine before shuffling (like a Combiner in MapReduce), greatly reducing network traffic.

PySpark Programming

PySpark: Word Count and Beyond

Classic Word Count in PySpark

```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("WordCount") \
    .getOrCreate()
sc = spark.sparkContext

# Load from HDFS
lines = sc.textFile(
    "hdfs:///data/corpus/*.txt")

# Transformations (lazy)
words = lines.flatMap(
    lambda l: l.split())
pairs = words.map(
    lambda w: (w, 1))
counts = pairs.reduceByKey(
    lambda a, b: a + b)

# Action: write result
counts.saveAsTextFile(
    "hdfs:///output/wordcount/")
```

bKash Fraud Detection Pipeline

```
# Load transaction records (Parquet)
txns = sc.textFile(
    "hdfs:///data/bkash/txns/")

# Parse CSV fields
def parse(line):
    f = line.split(",")
    return (f[0],          # txn_id
            f[1],          # sender
            float(f[2]))# amount

parsed = txns.map(parse)

# Flag suspicious: > 50000 BDT
# to new recipient in first txn
suspicious = parsed.filter(
    lambda r: r[2] > 50000.0)

# Count by sender
alerts = suspicious \
    .map(lambda r: (r[1], 1)) \
    .reduceByKey(lambda a, b: a+b) \
```


Spark SQL and DataFrames

Spark SQL and the DataFrame API

What is a DataFrame?

A **DataFrame** is a distributed collection of data organised into **named columns** with a schema — like a table in a relational database or a Pandas DataFrame, but distributed across a Spark cluster.

- Built on top of RDDs internally
- Uses the **Catalyst optimizer**: rewrites query plans for maximum efficiency (predicate pushdown, join reordering, constant folding)
- Supports both **SQL** syntax and **Python/Scala** method chaining
- Reads Parquet, ORC, CSV, JSON, Hive tables, JDBC

DGHS Health Analytics (Spark SQL)

```
# Load Parquet health records
df = spark.read.parquet(
    "hdfs:///data/dghs/records/")

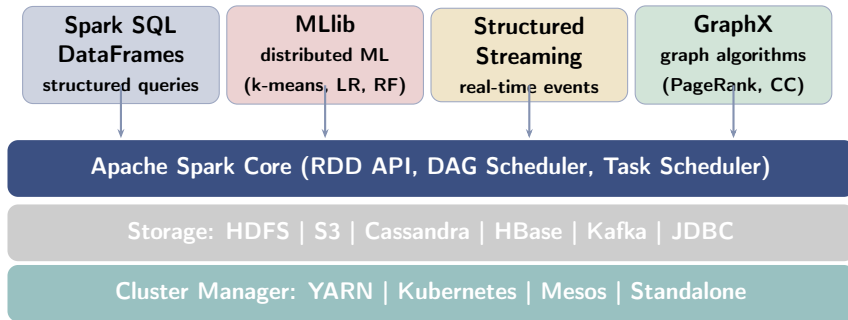
# Register as SQL view
df.createOrReplaceTempView(
    "health_records")

# Query: district-wise disease count
result = spark.sql("""
    SELECT district,
           disease_code,
           COUNT(*) AS cases
    FROM   health_records
    WHERE  year = 2024
           AND status = 'confirmed'
    GROUP BY district, disease_code
    ORDER BY cases DESC
    LIMIT 20
""")

# Write result as CSV
```

The Spark Ecosystem

Spark Ecosystem: One Engine, Many Workloads



MLlib: Distributed Machine Learning

Supports classification, regression, clustering, dimensionality reduction, and collaborative filtering. All algorithms are designed to run in-memory across the cluster, leveraging Spark's RDD caching for iterative training.

Structured Streaming

Treats a stream as a continuously-appended table; uses the same DataFrame/SQL API as batch. Supports event-time windowing, watermarks for late data, and exactly-once processing with Kafka.

MapReduce vs. Spark: Head-to-Head

MapReduce vs. Spark: Complete Comparison

Dimension	MapReduce	Spark
Intermediate storage	Disk (HDFS) after every step	Memory (spill to disk if needed)
Programming model	Map + Reduce only	map, filter, join, groupBy, SQL, ML, stream
Languages	Java (primary)	Python, Scala, Java, R
Iterative algorithms	Slow (disk per iteration)	Fast (cache RDD; 10–100×)
Fault tolerance	Re-execute task from HDFS	Recompute from lineage
Stream processing	Not supported natively	Structured Streaming (near real-time)
Interactive queries	Not feasible (>30 s)	Spark Shell / Zeppelin (sub-second)

Ivy League Alignment

Spark in Top University Curricula

Stanford CS246 (Leskovec) — Mining of Massive Datasets

The entire modern portion of CS246 is taught in **PySpark**:

- **RDD lineage** is covered as a formal directed acyclic graph (DAG) with typed edges (narrow vs. wide)
- **PageRank on Spark**: students implement the iterative algorithm and observe the $100\times$ speedup vs. MapReduce because the graph adjacency matrix is cached
- **Homework assignments**: all submitted as PySpark notebooks run on a shared Spark cluster; students must justify every `cache()` call with a cost estimate

Stanford exam question: *“Given the following Spark DAG, identify all stage boundaries and estimate the total shuffle volume for a 1 TB input dataset with 1,000 partitions.”*

MIT 6.830 — Database Systems

MIT frames Spark as a **query engine with deferred execution**:

- Catalyst optimizer = a relational algebra rewriter
- `DataFrame.explain()` shows the physical plan — same concept as `EXPLAIN` in SQL databases
- Spark SQL closes the gap between MapReduce-style systems and MPP databases

CMU 15-721 — Advanced DB Systems (Pavlo)

CMU benchmarks Spark SQL against

Student Assessment Pool

Instructions

Select the single best answer. Correct answers **bolded** for instructor reference.

Q1. What makes Spark significantly **faster than MapReduce** for iterative algorithms?

- ☐ a Spark uses more CPU cores per node
- ☐ b Spark writes intermediate results to faster SSDs
- ☒ c **Spark caches intermediate data as RDDs in memory**
- ☐ d Spark skips the shuffle phase entirely

Q2. How does Spark achieve **fault tolerance** in RDDs *without* full data replication?

- ☐ a Checkpointing to HDFS every 60 seconds
- ☒ b **Tracking lineage and recomputing lost partitions from parent RDDs**
- ☐ c Copying each RDD partition twice across two Executors
- ☐ d Using a write-ahead log on the Driver node

Instructor Notes

Q1: Option (a) is a common distractor — Spark uses the same hardware as MapReduce. The key is

Q3. Which of the following is **NOT** a component of the Spark ecosystem?

- ☐ (a) MLlib
- ☐ (b) GraphX
- ☐ (c) Structured Streaming
- ☐ (d) **HBase**

Q4. In Spark, a “transformation” such as `map()` or `filter()` is:

- ☐ (a) Immediately executed when called and materialised to memory
- ☐ (b) **Lazily evaluated — only executed when a subsequent action is called**
- ☐ (c) Automatically persisted to HDFS after execution
- ☐ (d) Only available in the Scala API, not in PySpark

Instructor Notes

Q3: HBase (option d) is a separate NoSQL database (covered in Week 4) that runs on HDFS. It can be used as a *data source* for Spark but is not part of the Spark ecosystem itself. **Q4:** This is the fundamental “lazy evaluation” concept. Students who answer (a) have a critical misconception that leads to inefficient Spark programs. Show `rdd.filter().count()` vs. just `rdd.filter()` to demonstrate.

Instructions

State True or False and provide a **one-sentence justification** for full marks.

Statement	Answer
1. Spark can only run on top of Hadoop YARN and has no other deployment option.	False
2. RDDs are mutable — individual elements can be modified in place after creation.	False
3. Spark's DataFrame API provides schema-aware, named-column operations similar to SQL tables.	True
4. MapReduce is generally better than Spark for machine learning workloads that require many iterations over the same dataset.	False

Instructions

Answer each question in **100–150 words**. Include a diagram where indicated.

- D1.** Explain the concept of a **Resilient Distributed Dataset (RDD)**. What does “*resilient*” mean in this context? How does Spark recover a lost RDD partition without replication?
- D2.** Differentiate between **Spark transformations** and **actions**. Give two concrete examples of each. Why does Spark not execute transformations immediately?
- D3.** Describe Spark’s **lazy evaluation model**. What is the advantage of deferring computation until an action is called? Use the example `rdd.filter(x > 1000).first()` to illustrate.

Instructions

Answer in **200–300 words** with step-by-step reasoning and quantified estimates.

A1. Iterative I/O Cost Analysis:

You need to train a K-Means clustering model over **50 GB** of bKash transaction data with **100 iterations**. Assume HDFS read speed = 500 MB/s and RAM cache read speed = 5,000 MB/s.

- Calculate the total HDFS I/O for MapReduce (all 100 iterations)
- Calculate the total I/O for Spark (50 GB fits in cluster RAM)
- Estimate the wall-clock time for each approach
- Under what conditions would Spark NOT outperform MapReduce?

A2. RDD Failure Recovery Trace:

A Spark job computes `input` $\rightarrow$ `filteredRDD` $\rightarrow$ `mappedRDD` $\rightarrow$ `reducedRDD` over 10 partitions. Mid-execution, the Executor holding partitions 3 and 7 of `reducedRDD` crashes. Trace the complete recovery sequence: what does the Driver detect, which partitions of which RDDs need to be recomputed, and what is the worst-case recovery cost if `reducedRDD` involves a **wide** dependency (`groupByKey`)?

Textbook & Reading References

Primary Textbooks for Week 6

- **[TW]** Tom White. *Hadoop: The Definitive Guide*, O'Reilly, 4th ed., 2015. **Ch. 2, 19**
 - Ch. 2: MapReduce (baseline for comparison) · Ch. 19: Spark (appendix)
- **[SA]** Acharya & Chellappan. *Big Data and Analytics*, Wiley, 2015. **Ch. 7**
 - Ch. 7: Introduction to Apache Spark
- **[LRU]** Leskovec, Rajaraman & Ullman. *Mining of Massive Datasets*, 3rd ed. **Ch. 2.4**
 - Ch. 2.4: Spark's model; extensions beyond MapReduce

Supplementary (Covered in Week 6, Part 2 — Seminal Papers)

- Zaharia, M. et al. (2010). **Spark: Cluster Computing with Working Sets**. *USENIX HotCloud 2010*.
- Zaharia, M. et al. (2012). **Resilient Distributed Datasets: A Fault-Tolerant Abstraction for In-Memory Cluster Computing**. *USENIX NSDI 2012*. **Best Paper Award**. **[PRIMARY SEMINAL PAPER]**
- Karau, H. et al. *Learning Spark*, O'Reilly, 2015 (1st ed. freely available as PDF).

Questions & Discussion

Week 6, Part 1 | CSE 4345 Big Data Analytics

“We introduced RDDs to provide a programming model that allows users to explicitly control how data is cached in memory across operations — a capability that simply did not exist in MapReduce.”

— Matei Zaharia, NSDI 2012

Next Lecture: Week 6, Part 2

Seminal Paper 1: Zaharia et al. (2010) — *“Spark: Cluster Computing with Working Sets”*

Seminal Paper 2: Zaharia et al. (2012) — *“Resilient Distributed Datasets”*

Case Study: Databricks logistic regression benchmark (2014) — 100× vs. Hadoop MapReduce on the same cluster

Reading before Part 2: [SA] Ch. 7 · Zaharia et al. (2012) RDD paper (see references)